awebster@falsecolour.com
+1 541 357 9546
Canyon, California (San Francisco Bay Area)
https://falsecolour.com/aw
github.com/AaronWebster

Summary Physicist turned embedded systems engineer who works across the hardware/software boundary. Over ten years at Google building the sensor platforms behind Street View: the IEEE 1588 time-synchronization stack, lossless multi-10 GbE capture data planes, and the MicroBlaze firmware and bring-up tooling on the camera FPGAs. Ph.D. in physics with first-author papers in *Nature Communications* and *Optics Letters* and a patented instrument. Reads Verilog, probes boards, ships production C++, and makes hardware debuggable by people and AI agents alike.

Strengths

- **Precision timing:** IEEE 1588 PTP, hardware timestamping, transparent clocks, clock modeling and interpolation (Kalman and RTS smoothing, PCHIP, integer-only FPGA arithmetic), PTP-scheduled triggering, TDEV and Allan-variance validation, TSN.
- **FPGA and embedded firmware:** AMD/Xilinx MicroBlaze on UltraScale+ and Kintex, AXI peripherals, Vivado, JTAG and IBERT bring-up, POST, watchdogs, A/B flash update over quad-SPI; STM32, ESP32, AVR.
- **High-rate Linux data planes:** `PACKET_RX_RING` (TPACKET_V3) with BPF, `SO_TXTIME`, zero-copy shared memory, multi-10 GbE capture; 10G PHYs, SFP+, MDIO Clause 45.
- **Hardware/software integration:** board bring-up, root-causing electrical faults, hardware-in-the-loop benches, Wireshark dissectors, Emboss binary-format code generation, Debian packaging.
- **AI-native engineering:** agentic coding workflows (Claude Code, Gemini CLI, Copilot agents), MCP tool servers, local LLM and Whisper inference on self-run GPU infrastructure.
- **Languages:** C, C++, Python, Bash (expert); TypeScript, MATLAB, Tcl, Lua, Starlark/Bazel, SQL, LaTeX (experienced); Verilog (reading).

Experience

Google LLC, Mountain View, California 2015–present

Software Engineer, Geo Hardware (Street View sensor platforms). Contractor 2015, full-time from February 2016, promoted 2017, tech lead since 2018. Software lead on the previous-generation capture system; embedded software lead for the FPGA team since 2024.

- **Time synchronization.** Wrote the PTP synchronization stack and algorithms used across the team's current systems: an FPGA offset estimator in unsigned-integer arithmetic, an $n \log n$ local-to-global time interpolator, a clock interpolator designed for $\sigma < 4$ ns over four hours (min-RTT anchoring, PCHIP), and PTP-scheduled triggering for a seven-camera array. Characterized clocks against a rubidium standard, Intel and Broadcom NICs, and PPS; offline oscillator analysis traced the cameras that kept failing to 5–10 ppm drift and drove a better oscillator onto the board respin.
- **Capture data plane.** Built the data plane, FPGA integration, and OS image of the previous-generation Street View capture system. Designed the next platform's data plane: lossless capture of multiple saturated 10 GbE links via zero-copy `PACKET_RX_RING` and BPF, message reassembly, self-describing packet headers with hierarchical drop counters, picosecond-timestamped frames published over shared memory with multicast announcement, and a pktgen testbed simulating multiple 10G streams.
- **Camera FPGA firmware (2024–present).** Own the MicroBlaze control plane of the next-generation vehicle camera: UDP command and logging, power-on self-test (DDR, PHY link, temperatures), watchdog, and A/B firmware upgrade over quad-SPI. Shipping production firmware releases since early 2026; validated PTP against a reference grandmaster with TDEV.
- **Bring-up and root cause.** Traced intermittent camera link failures to an under-specified crystal on the 10G PHY; found an AUX-board capacitor killing the UART; drove the 10G PHY mode decision with the silicon vendor's FAE; flagged an outsourced camera design as high risk early, before it was replaced by an in-house design. Took on electrical-engineering work after a reorg: board rework, custom SFP modules, a relay-automated hardware-in-the-loop bench for camera batches.
- **Developer tooling.** Built the team's camera bring-up and debug tool (C++ TUI and CLI over JTAG and UDP, self-test, bundled bitstreams, usage telemetry, Debian packaging), with an MCP interface for AI coding agents in development; Wireshark dissectors; Emboss-defined wire formats and register maps with generated C++ views.
- **Influence.** Internal talks on timelines and PTP (2023) and on "AI is the consumer" and actionable visibility (2026); about a dozen technical interviews a month.

Open source

- **google/emboss**, Google's binary-structure compiler for embedded C++: 25 merged pull requests, 2020–2026. Optimized generated C++ validation for structures with many conditional fields; added flooring division and modulus with undefined-value propagation; an embedded code-size benchmark

with a CI pull-request report; a clang-format back end; Bazel 9 and Arduino ESP32/ESP8266 support. In review: a Wireshark Lua dissector back end, a descriptor back end, and mypy typing.

- Personal repositories: autocovariance least-squares noise estimation; Intel igb PPS timing; a DKMS kernel-module patch exposing DualSense Edge controls; a real-time PCM to AC-3 (IEC 61937) encoder daemon; ChromeOS fleet automation for a K-12 school (Admin SDK, Chrome Policy API); Emboss definitions for USB-PD, UniFi, and cemuhook protocols.

Research

Max Planck Institute for the Science of Light, Erlangen, Germany 2011–2015

Doctoral researcher (Dr. Frank Vollmer). Invented the centrifugal force quartz crystal microbalance, a force-based biosensor that probes a sample's mechanical properties non-destructively in real time (*Nature Communications* 2014; PCT patent application). Discovered a new interference mechanism in conically scattered surface plasmon light and used plasmon speckle to resolve single-nanoparticle events in an unmodified SPR setup (*Optics Letters* 2013; *Nanotechnology* 2022). Built the optics, electronics, and data acquisition for each instrument; supervised a master's thesis.

Earlier research and engineering 2005–2011

FAU Erlangen-Nürnberg (2010–2011): parallel cluster DFT code for ultrashort pulses in high-NA optics. University of Oregon (2005–2010): plasmon scattering simulations; atom-optics instruments (scanning Michelson wavemeter at 1 in 10^6, low-noise photodetectors, external-cavity diode laser). Light Beam Industries, Eugene (2007–2009), senior R&D technician: fifteen documented PCBs for LED power, control, and thermal management (five in production, zero functional errors) and a one-meter integrating sphere.

Education

Dr. rer. nat. (Ph.D.), Physics 2018
FAU Erlangen-Nürnberg and Max Planck Institute for the Science of Light, Erlangen, Germany
Dissertation: *Novel Methods in Optical and Mechanical Biosensors*

Master of Science, Theoretical Physics: Optics 2011
FAU Erlangen-Nürnberg. Thesis: *Ultrashort Pulses in Focal Regions*

Master of Science, Applied Physics: Optics 2009
University of Oregon, Eugene, Oregon

Bachelor of Science, Physics 2007
University of Oregon, Eugene, Oregon

Publications

- Hongki Lee, Joel Berk, Aaron Webster, Donghyun Kim, and Matthew R. Foreman. "Label-free detection of single nanoparticles with disordered nanoisland surface plasmon sensor". In: *Nanotechnology* 33.16 (2022), p. 165502
- Aaron Webster, Frank Vollmer, and Yuki Sato. "Probing biomechanical properties with a centrifugal force quartz crystal microbalance". In: *Nature Communications* 5 (2014), p. 5284
- Aaron Webster and Frank Vollmer. "Interference of conically scattered light in surface plasmon resonance". In: *Optics Letters* 38.3 (2013), pp. 244–246
- Aaron Webster. "Novel Methods in Optical and Mechanical Biosensors". PhD thesis. Erlangen, Germany: Friedrich-Alexander-Universität Erlangen-Nürnberg, 2018

Patents

- Aaron Webster and Frank Vollmer. "Centrifugal force quartz crystal microbalance". Pat. req. WO 2015/042025 A1. PCT/US2014/055806, filed 16 September 2014. Mar. 26, 2015

Community

- **Community network, Canyon, CA** (2020–present). Co-founded a volunteer network sharing school fiber and wireless backhaul with neighbors: about 3,500 ft of volunteer-run fiber with fusion splicing and OTDR testing, point-to-point wireless links, per-household VLANs and failover.
- **Canyon Elementary School District** volunteer IT lead (2018–present): network, Google Workspace and ChromeOS fleet, servers, website. **Canyon Education Foundation** board member (2026–present).
- **Shared GPU and storage server** (2023–present). Co-built and administer a multi-GPU Linux server with a 58 TB ZFS pool for a small group of users: local LLM and Whisper inference, media services, Prometheus and Grafana, step-ca SSH certificates, and an OAuth identity-aware proxy.
- Amateur radio operator KM6MCS; coordinated repeater W6FMU.

Languages English (native), German (CEF B1, conversational), Spanish (CEF A2).